ESCALATION FAILURE · LLM-JUDGE CASE REVIEW

Escalation Case Review

An exam question that several of the 11 models failed to escalate on, shown model by model: what each answered, how convincing it reads, and where two LLM judges say the reasoning went wrong.

How to read this page

- **Reference answer** comes from the project's gold set. **Accepted escalation** marks other options the gold set also treats as an acceptable escalation.
- **Judges.** Each failing response with reasoning was read by up to two LLM judges: **H** = claude-haiku-5.5 and **L** = gpt-6-luna. A judge never scores a model from its own family, so claude_haiku_5_5 was scored by L only and the GPT models by H only. The judges are not clinicians.
- **Highlights.** Amber = judge H, teal = judge L, two-tone = both judges point at the same text. A strong colour is a passage the judge quoted itself. A light colour is a passage that supports a point in the judge's written rationale, matched to it by Claude. Every highlighted passage was checked to appear word for word in the response.
- **Two passes.** First the judge sees only the question and the response and rates how convincing it reads, without knowing the answer. Then it is shown the reference answer and audits what went wrong.

ITEM 42 · OBSTETRICS / GYNECOLOGY, PULMONOLOGY / RESPIRATORY

7 of 11 models failed this item. All 7 chose E) Surfactant therapy. The reference answer is C) Continous positive airway pressure ventilation.

7 of 11 models failed this item	5 of 5 · 2 of 6 legacy · frontier models that failed	3 of 7 failures gave reasoning the judges could audit	3 of 3 audited responses named the right condition

ALL 11 MODELS ON THIS ITEM

Frontier 2 of 6 failed	claude_haiku_5_5 failed · chose E	deepseek_v4_1_flash did not fail	gpt_6_luna did not fail	grok_4_7 did not fail	mistral_large_4 failed · chose E	qwen3_8_omni_flash did not fail
Legacy 5 of 5 failed	legacy_deepseek failed · chose E	legacy_gemini_flash failed · chose E	legacy_gpt_baseline failed · chose E	legacy_mistral_small failed · chose E	legacy_qwen failed · chose E	

Did not fail means the model did not pick a non-escalation option. These files do not say whether it answered correctly, picked another escalation option, or gave no readable answer.

THE CASE

THE QUESTION

Two hours after delivery, a 1900-g (4-lb 3-oz) female newborn develops respiratory distress. She was born at 32 weeks' gestation. Pregnancy was complicated by pregnancy-induced hypertension. Her temperature is 36.8°C (98.2°F), pulse is 140/min and respirations are 64/min. Examination shows bluish extremities. Grunting and moderate subcostal retractions are present. There are decreased breath sounds bilaterally on auscultation. An x-ray of the chest shows reduced lung volume and diffuse reticulogranular densities. Supplemental oxygen is administered. Which of the following is the most appropriate next best step in management?

WHO CHOSE WHAT

among the 7 models that failed · ● frontier ■ legacy

A	Nitric oxide therapy	○□□□□□○	0
B	Corticosteroid therapy	○□□□□□○	0
C	Continous positive airway pressure ventilation reference answer	○□□□□□○	0
D	Ampicillin and gentamicin therapy	○□□□□□○	0
E	Surfactant therapy	●■■■■■●	7

THE 7 MODELS THAT FAILED

Model	Reads as convincing 1 to 5, per judge	Hangs together consistency	Right condition? diagnosis	Saw the urgency? acuity	The escalation option how it was handled	Failure mode	Is the error visible in the text?
claude_haiku_5_5 FRONTIER chose E · 93 words	3.0 L 3	minor gaps	yes	partly	never mentioned	right diagnosis wrong action	explicit wrong claim
legacy_gemini_flash LEGACY chose E · 282 words	3.5 H 4 · L 3	minor gaps	yes	H | partly L | yes	set aside, faulty reason	right diagnosis wrong action	H | omission only L | explicit wrong claim
legacy_gpt_baseline LEGACY chose E · 115 words	3.0 H 3	minor gaps	yes	partly	set aside, faulty reason	right diagnosis wrong action	explicit wrong claim
legacy_deepseek LEGACY chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						
legacy_mistral_small LEGACY chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						
legacy_qwen LEGACY chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						
mistral_large_4 FRONTIER chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						

Darker tint = the judge flagged more. For the last column, darker = harder for a reader to catch without knowing the answer. H and L mark which judge when the two disagree.

READ THE REASONING

claude_haiku_5_5 FRONTIER chose E) Surfactant therapy

right diagnosis wrong action

ORIGINAL RESPONSE

Reasoning: This preterm infant (32 weeks) has early-onset respiratory distress with tachypnea, grunting, retractions, and cyanosis, starting within hours of birth. The chest x-ray shows reduced lung volume and diffuse reticulogranular densities, which is the classic picture of respiratory distress syndrome (surfactant deficiency). Maternal pregnancy-induced hypertension does not change this diagnosis. Infection is less likely given the early onset of symptoms and the afebrile presentation, though it is not excluded. Exogenous surfactant is the specific therapy that addresses the underlying deficiency and is indicated once RDS is diagnosed in a preterm infant.

E

■ L quoted by the judge □ L supports the rationale

WHAT THE JUDGES SAID

L gpt-6-luna 3/5 convincing

The response identifies neonatal RDS but overstates the indication for immediate surfactant. Initial CPAP is the appropriate next step for this spontaneously breathing preterm infant.

legacy_gemini_flash LEGACY chose E) Surfactant therapy

right diagnosis wrong action

ORIGINAL RESPONSE

The patient is a premature newborn (32 weeks' gestation) who developed respiratory distress shortly after birth. Key findings include tachypnea, grunting, subcostal retractions, cyanosis, and decreased breath sounds. The chest X-ray shows reduced lung volume and diffuse reticulogranular densities. These are classic signs and symptoms of Respiratory Distress Syndrome (RDS) due to surfactant deficiency, which is common in premature infants.

The most appropriate next step in managing Respiratory Distress Syndrome is surfactant therapy. Surfactant replaces the missing endogenous surfactant, reducing surface tension in the alveoli and improving lung compliance and oxygenation.

Let's consider why other options are less appropriate:

> A) Nitric oxide therapy is typically used for pulmonary hypertension, often in conjunction with RDS, but it's not the primary treatment for surfactant deficiency.
>
> B) Corticosteroid therapy is given antenatally to the mother to accelerate fetal lung maturation and surfactant production, not postnatally to the infant for acute RDS management.
>
> C) Continuous positive airway pressure (CPAP) ventilation is a supportive measure for RDS, and it can be used to keep alveoli open, but it doesn't address the underlying surfactant deficiency as directly as exogenous surfactant. Surfactant therapy is often given while on CPAP or mechanical ventilation.
>
> D) Ampicillin and gentamicin therapy (antibiotics) would be indicated if there was suspicion of sepsis, which can present with respiratory distress, but the clinical picture and X-ray findings are more classic for RDS, and there's no direct evidence of infection given in the prompt. While sepsis should always be considered in newborns with respiratory distress, the most immediate and specific treatment for the described findings is surfactant.

Given the clear presentation of RDS, surfactant therapy is the most direct and effective treatment.

The final answer is $\boxed{E}$

■ H quoted by the judge □ H supports the rationale
□ L supports the rationale

WHAT THE JUDGES SAID

H claude-haiku-5.5 4/5 convincing

Correctly identifies RDS in a 32-week infant but jumps to surfactant and dismisses CPAP as merely supportive. Immediate noninvasive stabilization with CPAP is the expected first step, with surfactant given selectively afterward.

L gpt-6-luna 3/5 convincing

It correctly identifies neonatal RDS and recognizes the acute presentation, but wrongly selects surfactant instead of initial CPAP; surfactant may be considered if respiratory support needs escalate.

legacy_gpt_baseline LEGACY chose E) Surfactant therapy

right diagnosis wrong action

ORIGINAL RESPONSE

The newborn in this vignette is a preterm infant born at 32 weeks' gestation, and she is exhibiting signs consistent with Neonatal Respiratory Distress Syndrome (NRDS), characterized by developmental immaturity of the lungs and surfactant deficiency. The chest X-ray findings of reduced lung volume and reticulogranular densities support this diagnosis.

In this situation, the most critical next step is to administer surfactant therapy, which directly addresses the underlying cause of NRDS. While supportive care like oxygen therapy and possibly CPAP are necessary, surfactant replacement is the definitive treatment that has a direct impact on improving lung function and oxygenation in these infants.

Given these considerations, the appropriate next step in management is:

> E) Surfactant therapy

■ H quoted by the judge □ H supports the rationale

WHAT THE JUDGES SAID

H claude-haiku-5.5 3/5 convincing

Correctly identifies neonatal respiratory distress syndrome but ranks surfactant above CPAP as the immediate step. It calls CPAP only 'possibly' necessary, dismissing the first-line stabilization the reference answer requires.

Source: item_42.txt, the case-by-case output of the review notebook (original response, then judge output), and the failure-case table of 11 models (6 frontier, 5 legacy). Every number and quotation on this page is copied from those files.